

**IITP Finance Club
Investing Wing
Sub-Coordinator Selection Assignment**

**The Quantamental Portfolio Challenge:
Multi-Cap Alpha Generation**

Submitted by

Chinmay Choudhury
2501ME29
chinmay_2501me29@iitp.ac.in

Portfolio Dashboard

• Rs 5,00,000 virtual capital

• 10 stocks

• 1-2 year horizon

• Risk Parity weighted

STOCK	CAP	WEIGHT %	₹ ALLOCATION	QUANT FACTOR	THESIS
ICICI Prudential AMC	Large	17.73%	Rs 88,647	Quality	Rising SIP adoption drives AUM; profit = % of AUM daily; asset-light model; 115% ROCE
Force Motors	Large	7.47%	Rs 37,345	Value + Quality	Loss to profit turnaround; OPM 2% to 16% expanding; BMW/Mercedes OEM adds premium margin; zero debt; P/E 25x
Garden Reach Shipbuilders	Large	7.82%	Rs 39,084	Quality + Momentum	Naval shipbuilding monopoly; 3-4yr order book locked; defence capex tailwind; OPM 3% to 11% expanding
Waaree Energies	Mid	9.13%	Rs 45,635	Quality + Growth	Largest non-Chinese solar module manufacturer; 500GW govt target; module production 7.1 to 12.6GW; P/E 20x
Tips Music	Mid	12.72%	Rs 63,612	Quality	30,000+ song catalogue; perpetual streaming royalties; zero incremental cost; asset-light model; 122% ROCE
CMPDIL	Mid	6.99%	Rs 34,964	Value + Quality	Govt PSU; Coal India as anchor client; 61% mining consultancy market share; zero debt
Shilchar Technologies	Small	6.47%	Rs 32,359	Quality + Momentum	Manufactures power transformers; renewable boom driving grid expansion and transformer demand surge; OPM 11% to 29%
Swaraj Engines	Small	13.37%	Rs 66,841	Quality + Value	Vast majority of engines supplied to Mahindra tractors; decade-stable 13-14% margins; 58.7% ROCE; 76% dividend payout; zero debt
Sika Interplant Systems	Small	5.75%	Rs 28,769	Quality	Engineering solutions for defence and aerospace; sticky qualified customers drive pricing power; OPM 8% to 21%; undiscovered with zero analyst coverage
Monarch Networth Capital	Small	12.55%	Rs 62,744	Value	Asset-light stock broker; demat boom from 40M to 185M accounts; OPM 17% to 69%; P/E 13x

Stock Rationales — Large Cap

ICICI Prudential AMC

I came across ICICI Pru AMC while looking at high-ROCE financials on Screener. What stood out immediately was the business model — they earn an expense ratio on AUM every single day, which means money just keeps flowing in as long as the markets aren't in freefall. They do mutual funds, PMS, AIFs, offshore advisory — so it's not a one-trick shop. India's SIP penetration is still nowhere close to what you'd see in the US or even China, so there's a long runway here. The 115% ROCE is not a fluke either; it's structurally explained by the fact that incremental AUM costs almost nothing to service. The obvious risk I'd worry about is a prolonged market crash — AUM falls, fees fall, both at the same time. That's the tail risk here.

Force Motors

Force Motors was actually a surprise pick for me. I was originally skeptical about autos but the OEM angle changed my view. They supply engines and axles to BMW and Mercedes for their Indian assembly — most people looking at Force Motors completely miss this because they only see the LCV/MUV business. The turnaround story is real: OPM went from 2% in FY21-22 (when they were losing money) to 16% now, which is a very different company from what you'd have seen three years ago. Debt is zero. P/E at 25x for a company still in the early innings of a margin recovery looks reasonable to me. The risk I'm watchful on is that LCV demand can soften fast if oil prices climb sharply — freight economics get worse and transporters stop buying trucks.

Garden Reach Shipbuilders

GRSE builds ships for the Indian Navy and Coast Guard — that's basically it. There's no real private competition in this space, and it would be very hard for any new entrant to get a defence shipbuilding licence in India. What gives me confidence is the 3-4 year order book that's already signed — meaning revenue for the next few years is largely predetermined, which is unusual for a manufacturing company. OPM has gone from 3% to 11% as they've gotten better at executing these large contracts. The one thing I flagged as a concern was debtor days climbing from 34 to 64 — the government pays slowly and that's putting pressure on working capital. Not alarming yet, but worth watching.

Stock Rationales — Mid Cap

Waaree Energies

Waaree is the largest non-Chinese solar PV module manufacturer in the world, which is a genuinely strong moat given how hard it is to build that kind of scale. They're also vertically integrated into EPC, so they're not just selling modules — they set up the whole plant. India's 500 GW renewable target is the structural tailwind here; this isn't demand that dries up based on quarterly sentiment. Production is going from 7.1 GW in FY25 to 12.6 GW in FY26, which is rapid scaling. Two risks I kept coming back to: EPC retention money creates a cashflow lag even when revenues look good, and Chinese manufacturers are pricing panels at \$0.10-0.12/watt versus Waaree's \$0.18-0.20 in export markets. That pricing gap is a real threat to international margins.

Tips Music

Tips Music was probably my favourite find in the whole screening process. They own 30,000+ songs outright — Bollywood, regional, Punjabi — plus film rights. Not licensed. Owned. That means every stream on Spotify, YouTube, JioSaavn pays them royalties forever, with no renegotiation clause. Because they're not producing new content to earn that revenue, the asset-light model kicks in hard — OPM has gone from 55% to 73% just as India's internet userbase grew. ROCE is 122%. The one thing that made me slightly uncomfortable was promoter holding dropping 10.8% over three years. That kind of steady decline can mean dilution, or it can mean insiders slowly selling. Worth watching closely.

CMPDIL

CMPDIL is technically a PSU under Coal India — they act as the technical advisory arm for the Ministry of Coal and hold roughly 61% share in India's mining consultancy space. The captive client relationship is unusual: when your only client is the government and that client controls the entire coal sector, revenue stability is essentially built-in. Zero debt, and debtor days have actually improved from 135 to 99 which tells you collections are getting better. The long-run structural risk is obvious — India will eventually move away from coal as the energy transition plays out. But on a 1-2 year horizon that's not a realistic threat, and I think the market undervalues the near-term earnings quality here.

Stock Rationales — Small Cap

Shilchar Technologies

Shilchar makes power and distribution transformers and exports to 25+ markets across 5 continents. The demand tailwind is pretty direct — every solar farm and wind park that gets built needs to connect to the grid, and that means transformers. OPM went from 11% to 29% because revenues grew much faster than their cost base, which is operating leverage playing out exactly as you'd expect. Zero debt throughout. The one thing I want to see confirmed over the next couple of years is whether the 50.7% ROCE holds — that's an impressive number but it needs to sustain to justify the thesis, it could be partly a one-year spike from the order surge.

Swaraj Engines

Swaraj Engines is almost exclusively a supplier to Mahindra's tractor division — 97% of revenue from one customer. That sounds like a red flag at first, but I'd argue it's actually a feature here, not a bug. Mahindra dominates Indian tractors, and Swaraj is essentially their captive engine supplier. What really stood out to me was the margin stability — 13-14% OPM for a decade straight. That's not a cyclical surge, that's a business with pricing power and cost discipline baked in over years. ROCE of 58.7%, zero debt, and they pay out 76% of profits as dividends. The risk is obvious: if Mahindra ever shifts volumes or changes suppliers, Swaraj has nowhere else to go at that scale.

Sika Interplant Systems

Sika Interplant was the most interesting discovery in this project. They do systems integration, interconnect solutions, MRO services and distribution for aerospace, defence and automotive — and they hold a Qualified Indian Offset Partner status with a government defence production licence. What that means practically is that once a defence customer has qualified and integrated their system, switching is incredibly costly and disruptive. That stickiness explains how OPM went from 8% to 21% over a decade without a single down year on margins. Market cap is only Rs 1,942 Cr and there's zero analyst coverage — I genuinely don't think the broader market has found this one yet.

Monarch Network Capital

Monarch Network covers broking, primary markets, mutual funds, insurance and financial planning — so it's not a pure-play broker, which helps diversify the revenue a bit. The macro tailwind is hard to argue with: India's demat account base went from 40 million to 185 million in five years. That's a massive retail investor wave entering markets, and a chunk of them will end up as Monarch customers. Because it's asset-light, OPM went from 17% to 69% in three years — the cost base barely moved while revenues scaled. P/E is 13x which is the cheapest in this portfolio. The catch is real though — broking is cyclical. In a proper bear market, trading volumes collapse and earnings can halve quickly.

Weighting Strategy

Method: Risk Parity (Inverse Volatility Weighting)

$$\text{Weight}(i) = \frac{\frac{1}{\sigma(i)}}{\sum \left(\frac{1}{\sigma(j)} \right)}$$

- $\sigma(i)$ is the standard deviation of daily percentage returns of stock i
- $\sum(1/\sigma(j))$ is the sum of inverse volatilities across all 10 stocks
- Data period: May 20, 2025 to May 20, 2026, covering 247 trading days

(Three recently-listed stocks — CMPDI, Shilchar Tech, and Sika Interplant — had limited price history of 32, 118, and 20 trading days respectively. Proxy substitution was rejected to preserve data integrity; σ was calculated using each stock's actual available history only.)

The idea is simple: stocks that move around less get a larger slice of the capital, because lower volatility usually means more predictable earnings. So instead of allocating equal rupees to each stock, you allocate equal risk. Each position ends up contributing roughly the same amount of volatility to the total portfolio.

Why Risk Parity over alternatives

- Conviction-weighted means you'd be guessing at future returns for each stock, and if you're wrong on even one or two, you've created a concentration problem you didn't intend
- Sector-neutral weighting just treats every stock the same regardless of how volatile or stable it actually is — Sika Interplant and Waaree would get equal capital despite being very different risk profiles

Acknowledged Limitation

I'll be honest — Risk Parity is a single-factor model and volatility alone doesn't tell you about business quality. Ideally you'd combine σ with ROCE, D/E and P/E into a multi-factor weighting model. I didn't do that here because it adds complexity that's harder to explain and verify. The trade-off was deliberate — I wanted something mathematically clean and defensible over something more sophisticated but harder to interrogate.

Portfolio Diversification

The portfolio covers 6 sectors — Financial Services, Defence, Energy, Automobile, Media and Mining. But the point I want to make is that having different sector labels isn't enough; the business models within each sector also need to be structurally different. I tried to ensure that, and the partial correlations section below addresses the cases where two stocks share a sector label.

Partial Correlations Acknowledged

Four pairs share a broad sector label, which I want to address directly rather than ignore.

- **ICICI Prudential AMC** and **Monarch Networth** are both in Financial Services but the revenue mechanics are completely different. ICICI AMC charges an expense ratio on AUM every day — so even if markets go sideways, they keep earning. Monarch earns per transaction. In a flat or dead market, ICICI AMC is still making money while the other isn't. Not the same risk profile at all.
- **GRSE** and **Sika Interplant** are both Defence, but GRSE builds full warships under multi-year contracts for the Navy. Sika does engineering systems, interconnect solutions and MRO — shorter cycles, different procurement channels, mostly aerospace and defence electronics. Very different order cadence and customer base.
- **Waaree** and **Shilchar** are both in Energy, but they sit at completely different points in the value chain. Waaree makes solar modules — generation side. Shilchar makes transformers — transmission side. Different buyers, different contracts, different dependencies. Both benefit from the renewable boom, but through different paths.
- **Force Motors** and **Swaraj Engines** both fall under Automobile but are structurally quite different. Force sells complete vehicles — LCVs, MUVs — to end customers. Swaraj supplies diesel engines to one buyer for one product. An end-consumer brand versus a captive component supplier.

Cap Mix

- 3 Large cap stocks — anchoring the portfolio with earnings stability and lower drawdown risk
- 3 Mid cap stocks — adding sector coverage and growth visibility without too much volatility
- 4 Small cap stocks — where I think the undiscovered alpha in this portfolio lives
- 9 of 10 holdings are domestic revenue-driven, carrying minimal exposure to global risk-off events, currency fluctuations or export demand cycles
- 8 of 10 holdings show expanding operating margins over the last 3 years, meaning diversification here is not just sectoral but qualitative

Macro Stress Test

STOCK	SCENARIO 1 — RBI RATE HIKE (75 BPS)		SCENARIO 2 — OIL SPIKE (\$130/BARREL)	
	VERDICT	REASON	VERDICT	REASON
ICICI Prudential AMC	Hurt	Equity markets sell off, AUM drops, fee income falls	Neutral	Pure financial services, no physical inputs
Force Motors	Mild Hurt	Higher EMIs soften LCV demand	Hurt	Diesel spike worsens LCV freight economics
Garden Reach Shipbuilders	Neutral	Govt contract earnings locked, P/E stable	Neutral	Govt cost-plus, naval demand unaffected
Waaree Energies	Neutral	Solar orders govt-backed, unaffected by rates	Benefit	Oil spike accelerates renewable energy transition
Tips Music	Neutral	Streaming royalties flow regardless of interest rates	Neutral	Pure IP, zero physical inputs
CMPDIL	Neutral	Govt cost-plus contracts, zero debt	Neutral	Cost-plus model, new mines beyond 1-2yr horizon
Shilchar Technologies	Neutral	Infrastructure orders government-backed	Benefit	Grid expansion accelerates, more transformer orders
Swaraj Engines	Neutral	Agri loans heavily subsidised, hike does not pass through	Mild Hurt	Diesel costs rise but tractor demand is monsoon-driven
Sika Interplant Systems	Neutral	Defence contracts locked, zero debt	Neutral	Defence pricing power absorbs input cost increases
Monarch Network Capital	Hurt	Trading volumes collapse in rate hike environment	Neutral	Broking revenue unrelated to oil prices

Stress Test Synthesis

SCENARIO 1 — RATE HIKE

A 75 bps rate hike by the RBI leaves 62.25% of portfolio weight unaffected. The reason is simple — the main damage route in a rate hike is rising debt costs hitting company profits, but 9 of 10 stocks here have D/E below 0.1, so there's almost no debt channel to transmit through. The 30.28% that gets hurt (ICICI AMC and Monarch) feel it through market sentiment falling, not through any actual operational damage. Force Motors at 7.47% gets mild softening as higher EMIs make LCV purchases a harder decision.

Two Rate Hike Transmission Channels

- Debt channel: if companies are leveraged, higher interest rates directly eat into profit. This portfolio is almost entirely bypassed here — 9 of 10 carry D/E below 0.1
- Valuation re-rating channel: when FD rates go up, investors dump expensive P/E stocks. But this only really bites stocks with shaky earnings. The government-contracted names in this portfolio — GRSE, CMPDIL, Sika — don't get P/E-compressed because their earnings aren't cyclically uncertain

SCENARIO 2 — OIL SPIKE

At \$130/barrel, 63.56% of portfolio weight stays neutral. What surprised me is that part of the portfolio actually benefits — a higher oil price accelerates renewable policy urgency.

- Waaree and Shilchar together (15.60%) directly benefit — an oil spike makes the renewable transition more urgent, which pushes more orders their way
- Force Motors (7.47%) gets hurt — diesel gets expensive, freight economics worsen, LCV demand softens
- Swaraj Engines (13.37%) is mildly hurt — higher diesel costs impact tractor running costs, but tractor demand is mainly driven by monsoon and crop cycles, so the damage is limited

Two Oil Spike Transmission Channels

- Input cost pressure: companies buying oil-derived inputs see costs rise
- End-demand softening: customers reduce purchases when fuel costs spike

Both channels are confined to 20.84% of portfolio weight combined.

Combined Portfolio Resilience

Looking at both scenarios together, 62-64% of portfolio weight holds up in either case. What I found most interesting when running through this is that the renewable energy exposure creates an actual tailwind in the oil shock scenario — not just protection. That wasn't accidental. I wanted some positions that don't just survive a bad macro environment but actively do better in it. The low-leverage construction meant the debt transmission channel barely applies here, which is the bigger picture takeaway.

Conclusion

Expected Return

Portfolio average sales growth of approximately 32% and profit growth of approximately 46% over 3 years significantly outpaces the Nifty 500 historical average of 12 to 15% annually. With active sector tailwinds in Waaree and Shilchar, early-stage margin expansion in Force Motors and Garden Reach, and low valuation entry points in Monarch at P/E 13x and Swaraj at P/E 23x, a conservative expected return of 18 to 25% over 18 months is estimated.

*Return estimated using: Price Appreciation $\approx$ Earnings Growth * P/E Change*

(Since portfolio P/E multiples are moderate ranging from 13x to 53x, significant investor sentiment-driven P/E compression is unlikely. Price appreciation is conservatively estimated at 60 to 70% of the portfolio's earnings growth rate, adjusted further downward for macro uncertainty and capital market sensitivity risk).

Top 3 Risks

- A sustained bear market is the biggest single threat. ICICI AMC and Monarch together are 30.28% of the portfolio and both earn based on market activity. Broking revenues historically fall 40%+ in sharp downturns — the 2020 crash is a useful reference point here.
- A bad monsoon hits Swaraj Engines hard — 13.37% of portfolio weight, almost entirely dependent on Mahindra tractor demand, which farmers drive based on crop expectations. In 2023 when there was a monsoon deficiency, tractor industry volumes dropped around 7%.
- Chinese solar dumping is a real risk for Waaree in export markets. Chinese manufacturers are pricing panels at \$0.10-0.12/watt while India's cost structure lands around \$0.18-0.20/watt. Waaree's domestic positioning is strong but if export margins get squeezed consistently it'll show up in the financials.

What I Learned

Honestly, I came into this with almost no real finance background — I knew what a stock was but terms like ROCE, OPM and D/E were things I had to look up at the start. What changed is that I stopped treating them as numbers and started understanding what they actually measure. ROCE clicked when I realised it tells you how efficiently a business converts capital into profit — Tips Music and ICICI AMC having 120%+ ROCE stopped being just impressive numbers and started making sense as a direct consequence of their asset-light model. The stress test was probably the most intellectually interesting part — I didn't expect going in that rate hike and oil shock would transmit through completely different channels, or that an oil crisis could actually help 15% of the portfolio rather than just hurt it. The thing I'd do differently is build a multi-factor weighting model that combines σ with ROCE and D/E — Risk Parity is clean but it doesn't account for business quality, which I only really understood after building it.

Appendix — Methodology

- Screener.in query: Market Cap > 500 AND ROCE > 15 AND D/E < 1 AND Sales growth 3yr > 12 AND Promoter > 40, resulting in 181 companies
- Cap buckets: Large cap > ₹ 20,000 Cr, Mid cap between ₹ 5,000 and ₹ 20,000 Cr, Small cap < ₹ 5,000 Cr
- Composite scoring: 40% weightage to ROCE, 30% to D/E safety and 30% to P/E valuation, each normalised on a 0 to 1 scale. Reduced from 181 stocks to 12 to 10 stocks ultimately.
- Risk Parity formula: $\text{Weight}(i) = (1/\sigma(i)) / \sum (1/\sigma(j))$